**Unit:** 6 Unit 9: Data Statistical Analysis
**Topic:** Box Whisker (Dispersion)

**Name:** ________________
**Date:** ____________

## Challenge (Habanero)

**Q1.** What is the primary purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To show the median and quartiles of a dataset
(C) To visualize the mode of a dataset
(D) To compare the range of multiple datasets
(E) To calculate the standard deviation of a dataset

**Q2.** A gamer's scores are: $12, 15, 18, 20, 22, 25, 30, 35, 40, 45$. What is the lower quartile ($Q_1$)?
(A) 15
(B) 18
(C) 20
(D) 22
(E) 25

**Q3.** In a box-and-whisker plot, what does the line inside the box represent?
(A) Mean
(B) Median
(C) Mode
(D) Range
(E) Interquartile range

**Q4.** The interquartile range (IQR) of a dataset is 15. If $Q_1 = 10$, what is $Q_3$?
(A) 20
(B) 25
(C) 30
(D) 35
(E) 40

**Q5.** A dataset has the following values: $2, 4, 6, 8, 10, 12, 1000$. Which value is an outlier?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 1000

**Q6.** What is the main difference between a box-and-whisker plot and a histogram?
(A) A box-and-whisker plot displays the mean, while a histogram displays the median
(B) A box-and-whisker plot displays the median and quartiles, while a histogram displays the distribution of data
(C) A box-and-whisker plot displays the mode, while a histogram displays the range
(D) A box-and-whisker plot displays the standard deviation, while a histogram displays the variance
(E) A box-and-whisker plot displays the interquartile range, while a histogram displays the mean absolute deviation

**Q7.** The upper quartile ($Q_3$) of a dataset is 25. If the interquartile range (IQR) is 10, what is $Q_1$?
(A) 10
(B) 15
(C) 20
(D) 22
(E) 30

**Q8.** A game developer collects data on the time it takes for players to complete a level. The times are: $5, 10, 12, 15, 18, 20, 22, 25, 30$. What is the interquartile range (IQR)?
(A) 5
(B) 8
(C) 10
(D) 12
(E) 15

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### Open Ended Questions (FRQ)

**Q9.** A dataset has the following values: $1, 2, 3, 4, 5, 6, 7, 8, 9, 100$. Construct a box-and-whisker plot and identify the quartiles, interquartile range, and any outliers. Show your work and explain your reasoning.

**Q10.** Compare and contrast the use of box-and-whisker plots and histograms in data analysis. Provide an example of when you would use each and explain the advantages and disadvantages of each. Be sure to include a visual representation of both types of plots.

**Chef's Tips**

- Tip 1: Emphasize the importance of understanding the context of the data when constructing and interpreting box-and-whisker plots.
- Tip 2: Encourage students to use technology, such as graphing calculators or statistical software, to construct and analyze box-and-whisker plots.
- Tip 3: Remind students to always check for outliers and to consider the limitations of box-and-whisker plots when interpreting the results.